Grade 12 Model Exam — Chemistry

Time allowed: 60 minutes. Choose the best answer.

1. Which gas turns limewater milky when it is bubbled through the solution for a few minutes?

A. O_2

B. CO_2

C. N_2

D. H_2

2. What is the pH of pure water at 25 °C?

A. 0 B. 1 C. 7
D. 14

3. Which statement about catalysts is correct?

A) They are used up B) They raise ΔH

C) They lower the activation energy D) They shift equilibrium

4. Which of the following best explains why ionic compounds conduct electricity when molten but not when solid?

(A) Their ions can move only when the solid is melted or dissolved in water, so charge can flow

(B) Electrons are free in the solid

(C) Molten compounds have no ions

(D) Heat makes new electrons

Answer key: 1-B, 2-C, 3-C, 4-A